



Integrating AI in Psychotherapy: An Investigation of Trust in Voicebot Therapists

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Abstract

The use of Artificial Intelligence (AI) applications, such as voicebots, holds potential to alleviate the current shortage in mental health services. However, it is unclear whether clients would place the same level of trust in these agents as they do in human therapists. This study aims to explore clients' trust in self-disclosure and privacy when interacting with a voicebot therapist. Additionally, it investigates the potential impact of incorporating explainable AI (XAI) on trust perceptions. In a pre-registered, between-subjects design (2x2: human vs. voicebot, explanation vs. no explanation) study, 564 participants were presented with a pre-recorded therapy simulation. Participants were asked whether they would trust the human or voicebot therapist and whether they would be willing to use it in the future. The results revealed no significant difference in trust between the voicebot and the human therapist. Furthermore, increased transparency through XAI enhanced trust in the voicebot's overall functionality and led to higher levels of trust in its privacy measures.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Empirical studies in HCI**;

Keywords

Artificial Intelligence, Mental Health, Psychotherapy, Trust, XAI

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1 Introduction

Significant issues in the accessibility of mental health provision in low-, middle, and even high-income countries have only recently been reported by the WHO [1]. The continuing plight results in a significant proportion of untreated mental health conditions across all countries. The increasing demand for improved accessibility has served as a driving force behind the recent exploration and integration of artificial intelligence (AI) applications in the fields of psychotherapy and psychiatry [2]. This flourishing field of AI applications encompasses a spectrum of innovations, including the development of virtual agents serving as therapists [3], the employment of social robots in healthcare setting to treat anxiety and depression in elderly individuals with dementia [4, 5], or aiding children with autism spectrum disorder in enhancing their socio-emotional and functional skills [6]. Notably, the evolving role of artificially intelligent virtual and robotic agents extends beyond providing support for fundamental aspects of mental well-being, such as emotional comfort and social interaction. These agents are progressively assuming responsibilities associated with high-level therapeutic interventions, a domain that traditionally fell under the purview of extensively trained proficient healthcare practitioners like psychotherapists [7].

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With the emergence of such AI-powered mental health applications, questions arise how users perceive these agents, and whether differences to human-human interaction can be observed. A particular focus here is on the role of trust, as trust is not only important in the adoption of AI applications [8], but also for enabling individuals to share sensitive personal information, thereby contributing to the success of therapeutic interventions [9]. Importantly, concerning trust, contradicting predictions can be made for AI-powered mental health applications. On the one hand, previous research found that people are at times more inclined to open up to artificial conversational agents because they offer the possibility to comfortably talk about embarrassing topic without being judged [10, 11]. On the other hand, previous research also suggests that users have raised privacy concerns when engaging with such agents [10] due to systems' lack of transparency. Based on these contradicting predictions, the central research question of this work is:

RQ: For the context of psychotherapy, how does the trust relationship between users and an AI-powered mental health application play out in comparison to a human-human relationship?

In addition, we also focus on how to counter limited system transparency through the addition of system explanations (XAI). Building on prior studies that demonstrated the correlation between enhanced transparency and explanations with increased trust in AI systems [12–15], our research posits that transparent, intelligible explanations can similarly foster trust within the realm of psychotherapy. To examine this claim, we investigate the effects of adding an explanation to a therapeutic voice bot. Our findings offer crucial insights for advancing dependable and effective AI applications in mental health interventions.

2 Theoretical Background

2.1 Trust in AI: Influence of Ontological Class on Trust in the Context of Psychotherapy

Advanced AI-powered applications promise to revolutionize the mental health sector. Therapeutic apps such as the chat-based Woebot [16] or Wysa's virtual therapist [17] are no longer limited to rather low-level mental health support but have entered high-level therapeutic interventions. Independent of any human interference, these applications respond to clients and patients. Moreover, most of the currently existing AI-based psychotherapeutic applications employ conversational interfaces that simulate the experience of interacting with a human therapist, encouraging users to delve into and reflect on their emotional states and cognitive processes [18]. Empirical research on specific instances of chatbots has yielded evidence suggesting that certain users may form a perceived human-like connection with the chatbot, which may be indicative of the emergence of a personal relationship analogous to transference [19]. Hopes are that these applications have the potential to reach underserved populations and counter the current shortage of qualified health professionals [20]. A qualitative study by Gültekin and Şahin [21] demonstrated a high acceptance among mental health professionals towards AI-based mental health services; however, they point out that they cannot replace clinicians. However, with AI technologies stepping into roles previously solely held by humans, questions arise how users perceive these agents, and whether differences to human-human interaction can be observed. To that end, it

is of paramount importance to investigate trust dynamics between clients and AI applications because trust is not only essential in the adoption of AI applications [8], but also for the success of the therapeutic intervention [9].

Comparing trust in AI versus human therapists is peculiarly interesting as previous theoretical and empirical work makes contradicting predictions. To that end, following predictions by the machine heuristic as part of Sundar's Modality-Agency-Interactivity-Navigability (MAIN) framework, we could assume that users prefer and trust AI-based virtual therapists more than their human counterparts [22]. In the machine heuristic, Sundar theorizes that humans employ common stereotypes about machines when interacting with them such as the perceptions that machines are non-judgmental, secure, and devoid of ideological biases [23]. Supporting these claims, Schützler and colleagues found that users were indeed more ready to disclose sensitive information (here: questions regarding their drinking behavior) to a machine with a lower social presence rather than another human being due to the decreased social desirability [24]. Concepts such as "calculative objectivity" [25], and "mechanical objectivity" [26] echo claims of the machine heuristic, suggesting the "technologically inflicted promise of mechanical neutrality" ([27], p. 181).

Moreover, for the context of mental health, such mechanical neutrality might also offer a space that is more inviting to self-disclosure due to greater perceived confidentiality of machines as compared to humans regarding intimate issues. According to Aktan and colleagues, one of the most important benefits is the ability for individuals to comfortably talk about embarrassing experiences [11]. Furthermore, their study revealed that most of the participants preferred AI-based psychotherapy. However, the majority answered the question who they would trust more with their personal data with "human psychotherapist" [11]. Contrary, Topol notes that in a psychotherapy session many people prefer to communicate with a machine rather than a human being and are more willing to trust the machine with more personal information [20]. Similarly, research pertaining to virtual agents has focused on their role in eliciting health-related information, including issues such as sexually transmitted diseases, depression, and other psychological challenges [28]. Considering previous findings and recognizing that the therapeutic relationship between therapist and client relies on mutual trust, reciprocity and unique human connection it might initially appear controversial [29, 30]. However, the literature presented here indicates the existence of a distinct form of trust associated with nonjudgmental and neutral behavior by machines. Therefore, for this work, we will describe this line of arguments which suggests an advantage of virtual therapists, in the present study a voicebot, over human therapists as trust to self-disclose. Thus, it remains unclear, if the preference for AI-based psychotherapy could be attributed to a form of trust, stemming from the ability to freely discuss embarrassing topics. On the basis of the arguments following the machine heuristic and findings on self-disclosure, we propose that:

H1a: Trust to self-disclose is higher when interacting with a voicebot compared to a human therapist (both explanation and no-explanation).

In contrast to such heightened trust to self-disclose, virtual therapists might also face disadvantages as compared to human

therapists. In particular, we argue here that the restriction of relationships, especially those involving technology, occurs when individuals are concerned about their personal privacy. Privacy apprehensions are linked with the illicit gathering, utilization, or management of personal information [31]. In the context of digital environments, privacy issues are generally classified into three types: (1) Territorial privacy, pertaining to physical surroundings, (2) privacy of a person against undue interference, and (3) information privacy, relating to control over the collection, storage, processing, and dissemination of personal data [32]. The examination of how users establish trust in technology is primarily rooted in HCI literature [33]. Additionally, perceived privacy concerns have been shown to have negative effects on individuals' trust formation. This is particularly relevant in the context of human interaction with Virtual Assistants (VAs) such as Amazon's *Echo*, Apple's *Siri*, or Microsoft's *Cortana*, given that users are aware of the privacy implications associated with the use of these devices [34]. More concretely, concerns regarding data privacy have been identified as obstacles to the continued and regular use of conversational agents, such as chatbots, especially when tasks involve using sensitive personal data, for example in health contexts [35, 36]. On the other hand, a study of Skjuve and colleagues [10] pointed out that privacy and information security are a driver of trust in a relationship with an AI system. In conclusion, the aforementioned literature is still lacking definite evidence regarding a differentiated evaluation of trust in privacy when using XAI in a voicebot. In this study, we will present arguments that suggest virtual therapists may have an inherent disadvantage compared to their human counterparts due to concerns about the trustworthiness of virtual platforms and the confidentiality of client information. In light of the preceding arguments pertaining to privacy, it is proposed that:

H1b: Trust in privacy is lower when interacting with a voicebot compared to a human therapist (both, explanation, and no-explanation).

2.2 Influence of Intelligible Explanations to Foster Trust in Psychotherapy

A range of human factors may exert a considerable influence on the level of trust placed in AI systems. These include user education about, past experiences with, and perception of automation, as well as the controllability, transparency, model compellability, and associated risks of AI systems themselves [37]. When designing an AI system, these factors need to be considered. Consequently, explainability and transparency are crucial factors in machine learning systems, with the level of importance varying depending on the application [38]. In critical domains like psychotherapy where high-stakes decisions are made, prioritizing explainability is paramount [12]. One possible solution is to provide an explanation of the AI (XAI), thereby increasing its perceived accountability and transparency, and ultimately enhancing trust in it [15]. XAI refers to machine learning and AI technologies that can offer humanly understandable justifications for their output or procedures [39]. XAI is divided into two categories: the global interpretation and the local interpretation, depending on the scope of the interpretation [38, 40]. In global XAI, the process is analyzed in a broader, more general, and goal-oriented way to understand the entire model. In

contrast, the local interpretation provides detailed explanations for each prediction and decision, clarifying why the model chooses a particular path. As the complexity of AI systems and algorithms grows, people increasingly regard them as "black boxes" that challenge comprehension in the sense that a large amount of skill and information are necessary to comprehend the AI decision or performance [41]. The presence of numerous complexities results in a lack of transparency that hinders understanding and negatively influences trust [12]. Additionally, explainability was identified as being crucial for building faith, rapport, trust, and bonding with AI [12–14]. Therefore, XAI in psychotherapy can facilitate the formation of a stronger therapeutic alliance with a virtual therapist by elucidating the rationale behind the selected treatment approach. Furthermore, it elucidates the variables that were taken into account during the development of the treatment plan [42]. This, in turn, enables patients to comprehend the mechanisms employed by the AI system, thereby facilitating informed decision-making with regard to their treatment. Consequently, XAI appears to positively impact trust in self-disclosure, leading to the following hypothesis:

H2a: Trust to self-disclose is higher when the voicebot is providing an explanation of its operating principles compared to when the voicebot provides no explanation.

The importance of system transparency in building trust in privacy and promoting a positive user experience has been proved in research [43, 44]. A study conducted by Weitz and colleagues found that the combination of XAI methods with linguistic information presented by a virtual agent presented visually can be beneficial for bringing trustworthy AI systems to end-users and contributing towards a responsible AI [13]. However, the study focused on the differentiation of XAI methods, namely XAI presented by a virtual agent or XAI without visual support. Additionally, only situational trust was examined here, which strongly depends on the context. Furthermore, the researchers point out that the effect could be attributed to the lifelike visual representation of the virtual agents [13]. Hence, while the findings offer valuable insights into the impact of XAI on trust, they do not address the distinction between a voicebot employing XAI and one that does not. Furthermore, a study conducted by Shin revealed that the presence of an explanation of the AI system increases its perceived accountability and transparency [15]. This, in turn, impedes comprehension and adversely affects trust in the system, since users may understand better what happens with their data and feel more secure sharing them. This is especially important in the context of psychotherapy, where patients must have a high level of trust in their therapist regarding their data security to effectively engage in therapy [45]. Additionally, a study by You and colleagues found that without upfront knowledge that participants' data was secure, participants were less willing to disclose information to a virtual agent [46]. This highlights the importance of data security and privacy in building trust between patients and auditory AI systems, thus emphasizing the critical role of providing patients with insights into the processing of their data. It remains unclear whether solely auditory XAI can engender the same effect as visually presented explanations. This warrants further examination, leading to the following hypothesis:

H2b: Trust in privacy is higher when the voicebot explains its operating principles compared to when the voicebot provides no explanation.

3 Method

Preregistration, data, and materials are available on the Open Science Framework (<https://t.ly/x7rQv>). Written confirmation from copyright holders was obtained for all images and the study was approved by ethics committee of the University Duisburg-Essen.

3.1 Experimental Design

As the conducted experiment employed a 2 (ontological class: human vs. voicebot) x 2 (explanation vs. no explanation) between-subjects design, four experimental conditions were tested within the questionnaire: human therapist without explanation of their professional background (condition 1), human therapist with explanation of their professional background (condition 2), virtual therapist (voicebot) without explanation of its operating principles (condition 3), and virtual therapist (voicebot) with explanation of its operating principles (condition 4), using a pre-recorded video stimuli.

3.2 Procedure and Stimulus Material

For practical reasons and ethical considerations relating to the sensitivity of the potentially disclosed data, it was decided to use video material as stimulated stimuli. Instead of placing the participants directly in a simulated therapy situation, they were asked to put themselves in the position of a therapy client using the video material provided. After an initial briefing, participants provided their informed consent. Subsequently, they received a short instruction, to inform the participants that they will be viewing a video depicting a simulated therapy session, specifically, an initial therapeutic consultation. To introduce the participants to a regular therapy setting, they were provided with a brief explanation that such session typically involves the patient and therapist acquainting themselves, with the patient articulation the purpose of their visit. Additionally, participants were informed that while a standard initial consultation typically lasts around 50 minutes, the video they were about to watch was only a 4-minute excerpt. They were instructed to watch it attentively and without interruption since the video couldn't be paused, and they had to respond to questions afterwards. Furthermore, participants were encouraged to empathize with the patient's perspective during the viewing.

Participants were randomly assigned to one of the four experimental conditions. In all conditions, participants watched a video depicting a human male patient, whose identity was concealed as he was only visible from behind, as seen in figure 1. This patient engaged in a conversation with either a human female therapist (conditions 1 and 2) or an artificial virtual therapist, which was represented as a voicebot on a smartphone (conditions 3 and 4). The voicebot's communication had been pre-recorded utilizing the text-to-speech tool "NaturalReader". Gender-matching text-to-speech was utilized to portray the voicebot as a female therapist, ensuring consistent representation across all conditions, mirroring the gender of the human therapist. As previously stated, the discussion presented was the inaugural session of a therapeutic consultation,

conducted in accordance with the tenets of cognitive-behavioral therapy (CBT). The dialogue between patient and therapist was uniform across all experimental conditions, commencing with the patient's brief mention of his motivation for seeking therapy, namely the experience of panic attacks. The therapist proceeded to inquire about the characteristics and frequency of these attacks. The video concluded with the therapist expressing her preliminary impressions to the patient.

In condition 3, which evaluated the impact of XAI, the voicebot provided an initial explanation at the beginning of the video, delineating its operational principles and outlining its utilization of AI, as well as its capacity to analyze patient responses through natural language processing. Additionally, the voicebot conveys that it continuously learns and evolves from every conversation but emphasizes that none of the information shared will be personally attributed to the patient. This can be considered as a more general type of explanation and categorized as global interpretation, which was chosen to ensure the comparability with the human therapist. In condition 2, human therapist explained her professional background at the beginning of the session, including the information that she is a trained psychologist specializing in CBT and has been in the profession for 10 years. Furthermore, she reassured the patient that all topics discussed are subject to strict medical confidentiality safeguards. After this, the conversation as described above followed. A detailed script of each conversation can be found in OSF.

Following the viewing of the video, participants were subjected to an attention and manipulation check. The attention check required participants to identify the psychological problem presented in the video, which was either panic attacks, depression or an eating disorder. Subsequently, a manipulation check was conducted to ascertain whether participants correctly identified the ontological class of the therapist depicted in the video, namely whether they perceived the therapist to be human or a voicebot. The second question was designed to ensure consistency in the application of XAI. Participants were prompted to verify specific statements according to their assigned condition: for the human condition, whether the therapist disclose their focus on CBT or not, and for the voicebot condition, whether the virtual therapist explained its use of NLP or not. Subsequently, participants completed several questionnaires. The first questionnaire assessed social trust in the therapist. Here, they were encouraged to empathize with the patient's perspective while answering the questions and were asked to imagine that they are currently undergoing therapy with the therapist (either human or virtual, depending on the assessed condition) shown in the video. This was followed by an assessment of the warmth and competence of the therapist. Afterwards, the participant's trust in privacy was assessed. This was accompanied by instructions that the questions refer to their general attitude towards the following statements, regardless of the video shown previously. Further elucidation on the measurement scales employed is outlined in chapter 3.3.

To supplement these scales, participants were asked about their general experience with voicebot's and their usage behavior. In addition, the experience with psychotherapy and digital tools in therapy was surveyed. Prior to providing their sociodemographic data (age, gender, education, occupation) at the end of the questionnaire, participants were informed that an AI-based application

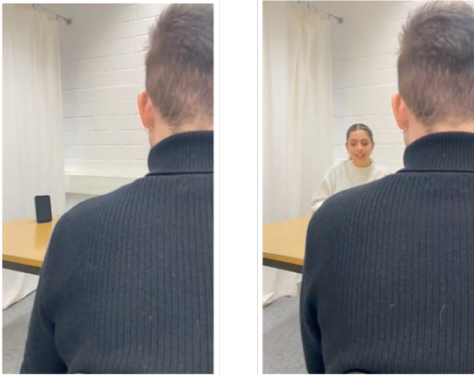


Figure 1: Video showing the virtual therapist (left) and the human therapist (right).

“PsyUDE” is currently being developed at the University Duisburg-Essen, which offers psychotherapeutic consultation with a voicebot via smartphone. Participants were given the opportunity to indicate their interest in participating in a trial phase of the app, which served as an indirect measure to elicit willingness to use a similar service as in conditions 3 and 4. Finally, the participants received a debriefing, which included anonymous contact services for mental health problems, and their compensation.

3.3 Sample

Participants were recruited and reimbursed via the panel provider ‘Prolific’. A total of 643 subjects were acquired. After the manipulation check, a total of 69 participants were excluded from the data set because they failed to answer the manipulation check correctly. Due to failure in attention checks, two participants were excluded, resulting in a final sample of $N = 572$. The surveyed sample achieves a power of .83 in the G*Power analysis with an effect of .15 and an alpha level of .05.

The remaining sample consisted of 277 men, 287 women, and 8 nonbinary individuals. The age ranged from 18 to 71 years ($M = 29.88$, $SD = 9.35$). The majority of participants were university students (40.2 %) or employees (50.5 %), holding a bachelor’s (28 %) or master’s degree (22 %). The use of voicebot’s within daily life was reported as never (36.9 %) or rare (33.6 %). Most of the participants never (42.3 %) or rarely (31.6 %) ask questions to a voicebot. Over half of the participants (67.8 %) had previously undergone therapeutic treatment, with 12.2 % currently receiving treatment, and a further 21.9 % intending to do so in the future. Of the respondents, 12.9 % had undergone a video chat therapy session, 6.3 % had utilized therapy apps, and 1.2 % had employed voice- or chatbots for therapeutic purposes. Finally, more than a third (37.2 %) expressed interest in the presented therapy app “PsyUDE”.

3.4 Measures

The following section contains an overview about the used measurements. All questions were answered on a five-point Likert scale ranging from 1 = “strongly disagree” to 5 = “strongly agree”.

3.4.1 Trust to Self-Disclose. Trust to self-disclose was measured by the subscale “Social Trust in Therapy (SV-TAB)” of the Short Scales to Assess Trust in Psychotherapy (VTT-TABs) [47]. The SV-TAB subscale is designed to measure confidence in the treating psychotherapist, consisting of 12 items, for example, “I can really tell my therapist everything” and “I know that I can tell my therapist anything without being rejected” ($\alpha = .88$). Depending on which video was presented to the subjects, they received slightly different instructions, referring either to a human or virtual therapist.

3.4.2 Trust in Privacy. Trust in privacy was measured by the Internet Privacy Concerns (IPC) scale by Hong & Thong [48]. This measure aimed to assess the general trust in privacy of participants, regardless of their assignment to either the voicebot or the human therapist condition. Since not all subscales of the IPC are appropriate for the intended study context of healthcare, only the following subscales were used for the survey: Secondary Usage, Improper Access, Trusting Beliefs, and Risk Beliefs (14 items, $\alpha = .75$). Moreover, the original scale refers to “commercial/government websites”, which was not suitable for the present context (healthcare). Therefore, the wording was changed to “digital health providers”, which were defined as websites or apps that store personal data such as name, address, and more sensitive medical data such as medical documents or diagnoses, resulting in statements like “I am concerned that digital health providers do not devote enough time and effort to preventing unauthorized access to my personal information”. As no validated German version of the IPC scale is available, the items were translated via translation and back translation procedure.

3.4.3 Covariates. We included *warmth and competence* [49] (warmth, $\alpha = .90$ and competence, $\alpha = .94$), as well as *attitude towards psychotherapy*, measured by the Attitude towards Seeking Professional Psychological Help Scale-Short Form (ATSPPH-SF) [50] ($\alpha = .11$ with the current sample), and *attitude towards AI*, measured by the German version of the ATAI scale [51] (acceptance, $\alpha = .65$ and fear $\alpha = .66$), as covariates.

4 Results

The descriptive statistics for all conditions are presented in Table 1. Differences displayed in trust to self-disclose and trust in privacy were tested by a 2 (human vs. voicebot) $\times$ 2 (explanation vs. no explanation) between-subjects MANCOVA.

4.1 Hypothesis 1: Influence of Ontological Class on Trust

In H1a, we proposed that trust to self-disclose would be higher when interacting with a voicebot compared to a human therapist, while H1b assumed lower trust in privacy when interacting with a voicebot therapist compared to a human therapist. Results of the multivariate analysis of covariance (MANCOVA) indicate that there was a statistically significant difference between the four experimental groups on the combined dependent variables (i.e., trust to self-disclose and trust in privacy), $F(6, 1128) = 2.576$, $p = .018$, partial $\eta^2 = .014$, Wilks’ $\Lambda = .973$. Post-hoc univariate ANCOVAs were conducted. A main effect of ontological class on trust so self-disclose

Table 1: Descriptive Statistics on Trust to Self-Disclose and Trust in Privacy

Condition	M	SD	n
Trust to self-disclose			
Human	3.54	.34	150
Human (explained)	3.57	.30	145
Voicebot	3.51	.40	122
Voicebot (explained)	3.54	.42	155
Total	3.54	.37	572
Trust in privacy			
Human	3.21	.51	150
Human (explained)	3.15	.48	145
Voicebot	3.21	.49	122
Voicebot (explained)	3.26	.51	155
Total	3.21	.50	572

was found. Results show a statistically significant difference between groups for trust to self-disclose, $F(3, 565) = 4.107, p = .007$, partial $\eta^2 = .012$. Pairwise comparisons with regards to trust to self-disclose reveal a difference between the human therapist without explanation of professional background and the voicebot therapist, ($p = .023, M_{\text{Diff}} = -.117, 95\text{-CI}[-.224, -.010]$) and between the human therapist without explanation of professional background and the voicebot therapist explaining its operating principles ($M = 3.54, SD = .42, p = .028, M_{\text{Diff}} = -.106, 95\text{-CI}[-.205, -.007]$). Scoring of trust to self-disclose is slightly higher in the human therapist condition without explanations of professional background ($M = 3.54, SD = .34$) compared to the voicebot therapist condition ($M = 3.51, SD = .40$). Thus, the hypothesis H1a is rejected. Furthermore, the post-hoc ANCOVA yielded no significant difference between groups for trust in privacy, $F(3, 565) = .841, p = .472$, partial $\eta^2 = .004$. Thus, H1b is rejected.

4.2 Hypothesis 2: Influence of Intelligible Explanations on Trust

Hypothesis H2a assumed a higher degree of trust to self-disclose through the incorporation of the voicebot's operating principles. As mentioned before, the MANCOVA yielded a statistically significant difference for all four experimental groups. Nevertheless, post-hoc pairwise comparisons concerning trust to self-disclose yielded no statistically significant distinction between the voicebot therapist and the voicebot therapist incorporating XAI ($p = 1, M_{\text{Diff}} = .011, 95\text{-CI}[-.092, .114]$). Therefore, H2a is rejected. Hypothesis H2b predicted that trust in privacy would be higher for a voicebot explaining its operating principles compared to a voicebot without explanation. Post-hoc pairwise comparisons with regards to trust in privacy reveal no statistically significant difference between the voicebot therapist and the voicebot therapist explaining its operating principles, ($p = 1, M_{\text{Diff}} = -.040, 95\text{-CI}[-.200, .120]$). Thus, H2b is rejected.

4.3 Covariates

The covariate *attitude towards psychotherapy* exhibited no statistically significant effect in our analysis. *Attitude towards AI* showed significant results regarding the effect on trust in privacy, $F(1, 565)$

$= 6.899, p = .009$, partial $\eta^2 = .012$, as well as *warmth and competence* regarding the effect on trust to self-disclose, $F(1, 565) = 186.109, p < .001$, partial $\eta^2 = .284$. To better understand how these effects came into being, exploratory mediations for both covariates were calculated. *Attitude towards psychotherapy* and *warmth and competence* do not mediate trust to self-disclose or trust in privacy.

5 Discussion

This study examines trust in psychotherapy, specifically between users and voicebot therapists compared to human therapists. In the following discussion, we undertake a comprehensive synthesis of our findings, integrating them into the broader discourse surrounding the adoption of AI technologies in the domain of mental health. The initial phase of our investigation focused on assessing the impact of the ontological class (human vs. voicebot) on two distinct dimensions of trust: trust to self-disclose and trust in privacy.

Although the study yielded some significant findings, the observed differences are marginal. Consequently, the results and subsequent discussion should be considered exploratory, with the understanding that further research is necessary to reliably assess the tested dimensions of trust. However, the results revealed a deviation from our hypothesis: Trust to self-disclose did not exhibit a significant increase during the displayed interaction with a voicebot, not even when its operating principles were explained. Instead, it was higher when participants interacted with a human therapist without explanation of their background. This suggests that participants may prefer interactions with human therapists without specific explanations of the professional background, as this is perceived as more authentic, and trust is less influenced by details of background and more by personal empathy and understanding. Additionally, it is important to note that in real-life therapy sessions, human therapists typically do not explain the treatment plan or reasons behind certain questions they ask during a session. This suggests that excessive explanations in human-human interaction might impede the therapeutic process, as the results indicate, patients tend to prefer less explanation.

However, the observation related to human-voicebot interaction contradicts earlier research, which suggested that individuals are

more willing to trust a machine with sensitive or embarrassing information than a human [20]. In this context, it is noteworthy that our survey instrument did not directly encompass the evaluation of participants' perceptions regarding the advantages of utilizing a voicebot as a therapeutic entity. Furthermore, the voicebot's answers might have been scripted too human-like, which could have led participants to not perceive the voicebot as a machine with a lower social presence, leading to a social desirability [22] and therefore undermining the machines' advantages. Nevertheless, the ontological class's effect on trust to self-disclose was found to be small in this study. This implies that participants perceive only a minimal difference between disclosing potentially sensitive experiences and personal information to a voicebot compared to a human, indicating a potential acceptance of such interactions in mental health care services. Surprisingly, trust in privacy is not lower when imagining an interaction with the voicebot. In fact, there is no significant difference of trust in privacy in the human therapist condition compared to the voicebot therapist condition. This could imply that there are no data security or privacy concerns as an obstacle for using AI based psychotherapy tools, or in combination with the previously mentioned results, that the manipulation failed to induce the participants to genuinely perceive their interaction as with an AI-based system.

Another factor investigated was explainability. Contrary to our hypothesis, the implementation of explanations in a voicebot therapist does not lead to increased trust to self-disclose (H2a) or trust in privacy (H2b), which might imply that the implementation of XAI was insufficient to create a meaningful change in people's overall view of the voicebot therapist. However, it is important to acknowledge that the comparability of the explanations may influence the perceived trust regarding self-disclosure and privacy. The explanations provided within the study (professional vs. technical) are not identical: in the human condition, the explanation highlighted the therapist's expertise, while in the voicebot condition it focused solely on technical details. Since the concept of XAI pertains to machine learning and AI technologies [39], this concept cannot be directly applied to interactions with humans. The explanation of the human therapist's professional background and expertise was selected because it may be perceived as akin to the operating principles of a voicebot. The limited comparability of the explanations could have affected the level of trust in the system's competence. However, this was not assessed, and might not have been a critical factor for the present study, as the focus was on trust in privacy and self-disclosure.

In the future, the explanations could be better tailored to the specific psychotherapeutic context and additionally emphasize aspects useful for this purpose, such as the machine's non-judgmental character. The implementation of explanations of operating principles seemed not to help enhance participants' trust in the privacy of their data. This might be due to the fact that in various circumstances, different users require various forms of explanation [38]. The interpretations provided in the explainability condition may have been too global, which left participants with pending questions, for example on the technical processes of a specific decision or the acquisition and storage of data. A local XAI may be more beneficial for patients in this context, enabling them to comprehend their unique situation. Additionally, voicebots, in contrast to human

therapists, may elicit heightened disclosure tendencies through reciprocal sharing. However, this could potentially be constructed as deceitful, highlighting the paramount importance of XAI within this domain. Lee et al. [52] have previously demonstrated that the inclination to disclose during therapeutic interactions with virtual agents is positively influenced when the agent reciprocates by disclosing information. Chatbots' self-disclosure can trigger humans to disclose their personal feelings and thoughts [53]. However, its implementation within a scientific study while safeguarding data privacy poses a notable challenge. To figure out what kind of explanations are needed in AI-based psychotherapy tools, it is necessary to engage further with potential users of these systems. This might help to boost users' faith in the system, and hence in the virtual therapist in general.

6 Limitations and Future directions

While this study contributes valuable insights into the conceptualization of trust and explainability in mental health interventions, there are several limitations that need to be addressed in future research. Firstly, as discussed earlier, the comparability of the explanations provided in the human and voicebot conditions might have impacted the perception of trust. Additionally, using pre-recorded stimuli rather than real-time interactions in a laboratory setting reduces the ecological validity of the findings. Participants might not engage with the scenario as fully as they would in a genuine therapy session. Furthermore, the participant's ability to imagine interacting with either a human or artificial therapist could also affect the results. Therefore, it is recommended that the study should be replicated in a more realistic setting to address these issues, while also considering ethical constraints. Moreover, the translation of the questionnaires may have introduced some degree of error or bias, as the low Cronbach's Alpha of the ATSPPH-SF scale indicates. Additionally, alternative methods of measuring trust, such as the commonly used reference of trust in psychotherapeutic relationships, may need to be considered. Furthermore, the study did not fully examine the influence of the XAI condition on trust in privacy. A closer linking of the concepts of explainability and privacy may be necessary to identify the impact of the XAI condition on privacy-related trust. For instance, providing more specific and transparent information about data storage and handling may increase trust in the voicebot's ability to protect privacy. Thus, it is possible that the measurement instrument used in this study did not fully capture the participants' trust in the voicebot's privacy protection. Future studies would benefit from a more specific instrument that directly addresses the privacy concerns of voicebot users. Overall, future research should aim to address these limitations by exploring the factors that influence trust and explainability in mental health interventions in clinical samples with diverse psychological problems. Larger samples and replications of the study should be used to confirm the reliability of the findings. These steps will lead to a more precise and accurate understanding of trust and explainability in mental health interventions with AI.

7 Conclusion

A number of studies investigating the potential applications of AI-systems in psychotherapy. However, none of these addressed the

potential conflict between two types of trust when combined with XAI methods in the context of virtual therapists. This study elucidates the role of the ontological class of a therapist and delineates the operational principles underlying the trust between therapist and client. The findings indicate that AI-based virtual agents have potential for effective utilization in contexts requiring trust, particularly in the domain of mental health. Privacy concerns regarding sensitive data was not identified as obstacle to the utilization of voicebot therapists. The potential for AI-based psychotherapy to address the shortage of therapy places and improve accessibility is promising. Nevertheless, further research is required to examine the complex relationship between explainability and privacy, as well as potential conflict between trust to self-disclose and privacy concerns. It is therefore imperative to develop more effective methods for assessing trust in AI systems within a therapeutic context, differentiating it from the human therapeutic relationship.

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